08/2/2024 FN

Hall Ticket Number:

MR-22

Code: B324CM2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular Examinations, Jan/Feb-2024

Software Engineering

(Computer Science and Engineering (AI & ML))

Time: 3 Hours

Answer ALL Questions

Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define software engineering and explain.	1	1	1	1
1. b)	List the types of software Myths?	1	4	1	1
1. c)	What is feasibility study?	1	1	2	1
1. d)	What is Requirement Validation?	1	1	2	10
1. e)	Define Software Architecture?	1	1	3	3
1. f)	What are called Static diagrams?	1	1	3	3
1. g)	What is validation testing?	1	1	4	9
1. h)	Why do we need to test software?	1	1	4	9
1. i)	Write about quality concepts?	1	2	5	1
1. j)	Distinguish between reactive and proactive risk strategies?	1	4	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	With a neat sketch, explain Capability Maturity Model Integration (CMMI).	5	3	1	1
2. b)	Compare and contrast different software process models.	5	2	1	1
	OR				
2. c)	Explain the evolving role of software?	5	2	1	1
2. d)	What are the generic process framework activities? Explain.	5	1	1	1
3. a)	What is the goal of requirements analysis phase? Why the requirements analysis phase is a difficult one? Justify your answer	5	3	2	2
3. b)	What are the desirable characteristics of a good software requirement specification document?	5	4	2	2
	OR				
3. c)	Explain the differences between functional requirements and non-functional requirements by giving suitable examples.	5	3	2	2
3. d)	Discuss how feasibility studies are important in requirement engineering process?	5	4	2	2
4. a)	Draw the class diagram and explain various relationships that exist between classes.	5	2	3	3
4. b)	Distinguish between sequence and collaboration diagrams.	5	2	3	3
	OR				
4. c)	What are the design principles of a good software design?	5	1	3	3
4. d)	Draw and explain Use case diagrams and its development process with suitable illustrations.	5	2	3	3
5. a)	What is software quality? What are the metrics used for source code analysis?	5	1	4	1
5. b)	What are the various testing strategies used for testing conventional software?	5	1	4	9
	OR				

P.T.O.

5. c)	What is white box testing? How white box testing is carried out? Demonstrate with example.	5	1	4	9
5. d)	Discuss about the metrics used for software Quality model.	5	4	4	1
6. a)	Discuss the role of formal technical reviews in achieving good quality software.	5	4	5	10
6. b)	What is risk identification? Discuss various methods used for risk identification.	5	1	5	1
	OR				
6. c)	What is Statistical Software Quality Assurance? Discuss its objectives.	5	1	5	1
6. d)	Explain ISO 9000 Quality standards.	5	3	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome